RAVINDRA KUMAWAT

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EDUCATION

JECRC UNIVERSITY **October 2021 - May 2025**
Bachelor's, Computer Science and Engineering (with Cloud Computing AWS) *GPA: 7.1*

PROFESSIONAL EXPERIENCE

Capgemini **Bengaluru (India)**
Software Engineer *Aug 2025 - Present*

- Software Engineer with experience building **AI-powered applications, data products, and cloud-native solutions**.
- Enjoy solving problems at the intersection of **LLMs, data engineering, workflow orchestration, and analytics**.
- Passionate about designing systems that balance **AI-driven intelligence with reliability, transparency, and user control**.

SKILLS

Technologies: Python, LangChain, Ollama (local model inference), DuckDB (in-memory SQL), Streamlit, standard ML/serialization libs (numpy/pandas) , Prompt Engineering, Tool Calling, AI.
Familiar: AWS, Azure,CrewAI ,RAG

PROJECTS

SecureCodeAI - Multi-Agent Security Analysis Platform

- **Architected a comprehensive security analysis platform** using CrewAI's multi-agent framework to perform automated static code analysis, secret detection, and vulnerability assessment across codebases.
- **Implemented intelligent LLM fallback architecture** supporting both local LM Studio/Ollama models and cloud-based Gemini API, ensuring flexible deployment options while maintaining cost efficiency and data privacy.
- **Engineered specialized security agents** for targeted analysis including secret scanning (API keys, tokens, credentials), SAST pattern detection (eval (), os.system(), unsafe subprocess calls), and vulnerability report synthesis with precise file path and line number mapping. Implemented **JWT-based authentication** with RBAC, enhancing security and user segmentation across the platform.
- **Optimized performance through bootstrap file gathering** that eliminates tool-call dependencies, reducing analysis time by 40% and improving reliability with smaller language models through direct codebase injection.
- **Built robust CLI interface** with structured JSON output, supporting batch processing, custom output paths, and comprehensive error handling with proper exit codes for CI/CD pipeline integration. Conducted **end-to-end testing** using **Selenium**, validating UI/
  across critical modules.
- **Leveraged Pydantic validation and few-shot prompts** to ensure consistent, accurate security findings with confidence scoring, while preventing false positives through strict evidence-based reporting protocols.

DataPilot- NLP -> Data Analysis

- Developed an **AI-powered Natural Language to SQL Analytics Copilot** using **LangGraph, Streamlit, DuckDB, and LLMs**, enabling business users to analyze CSV data through conversational queries.
- Architected a **multi-stage workflow engine** for **intent classification, SQL generation, SQL validation, query execution, insight extraction, and visualization recommendation**.
- Implemented **human-in-the-loop validation** allowing users to review and edit AI-generated SQL before execution, improving transparency and user trust.
- Built an interactive **Streamlit frontend** supporting multi-turn conversations and file uploads, delivering a seamless and intuitive user experience.
- Developed a **Row Intelligence module** that analyzes query results, extracts business insights, and recommends appropriate visualizations based on returned data structures.

SafePaste – Secure Clipboard Secret Manager | Python, CLI

- Developed **SafePaste**, a lightweight **CLI utility** for securely copying sensitive information such as **API keys, tokens, passwords, and environment variables** to the system clipboard.
- Built a flexible **secret resolution engine** capable of retrieving values from **environment variables, files, and .env configurations** through a unified interface.
- Implemented native **Windows Clipboard Integration** using the **Win32 API and ctypes**, enabling direct interaction with system clipboard memory for secure data handling.
- Engineered an **automatic clipboard clearing mechanism** that removes copied secrets after a configurable timeout, minimizing the risk of accidental credential exposure
- Developed a **Row Intelligence module** that analyzes query results, extracts business insights, and recommends appropriate visualizations based on returned data structures.

ACHIEVEMENTS

- **AWS Developer Associate**
- **AWS Certified Cloud Practitioner**
- AZ-900